AI-Powered Credit Risk Intelligence Platform

Home Credit Default Risk | 307,511 applications

Not just a model — a credit decision system:

scored, explained, defensible, auditable, and queryable in plain English.

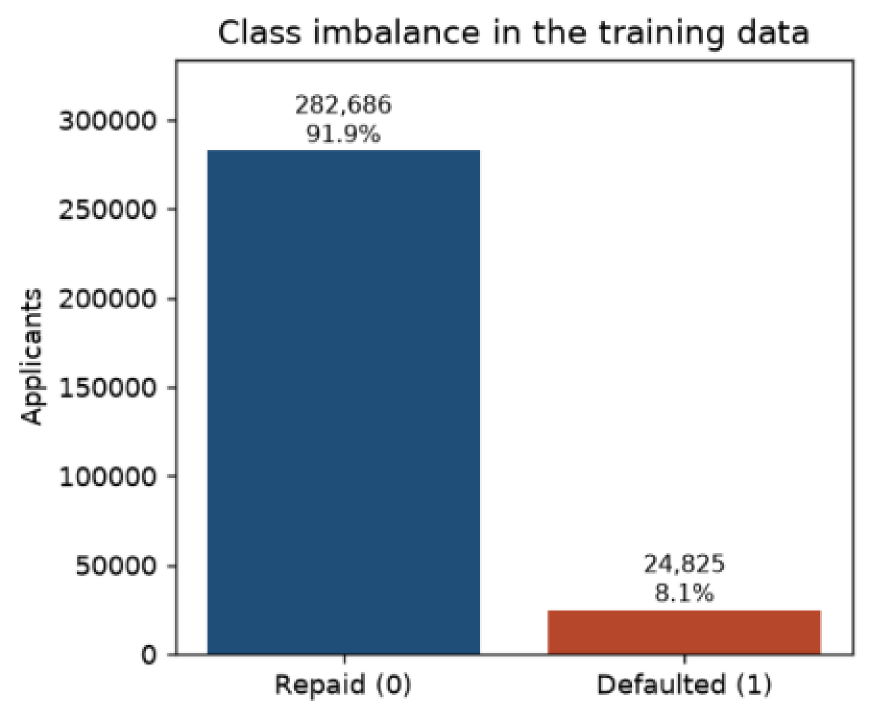
0.779	**0.264**	**0.0039**	**$221.39M**
Holdout ROC-AUC	PR-AUC at an 8% base rate	Calibration error (from 0.2687)	Expected loss avoided

Every figure in this deck is read from reports/ at build time.

The problem, and what makes it hard

A lender must decide, price and justify — in that order

- Predict default on 307,511 loan applications, 122 raw columns, 8.07% default rate (11.4 : 1 imbalance).
- Accuracy is a trap: predicting ‘everyone repays’ scores 91.9%. PR-AUC and expected loss are the metrics that mean anything.
- A score is not a decision. The threshold, not the model, determines how much money the portfolio loses.
- A decision that cannot be explained cannot be issued: ECOA / Regulation B requires the principal reasons for every decline.
- Analysts ask questions faster than a BI backlog can answer them.

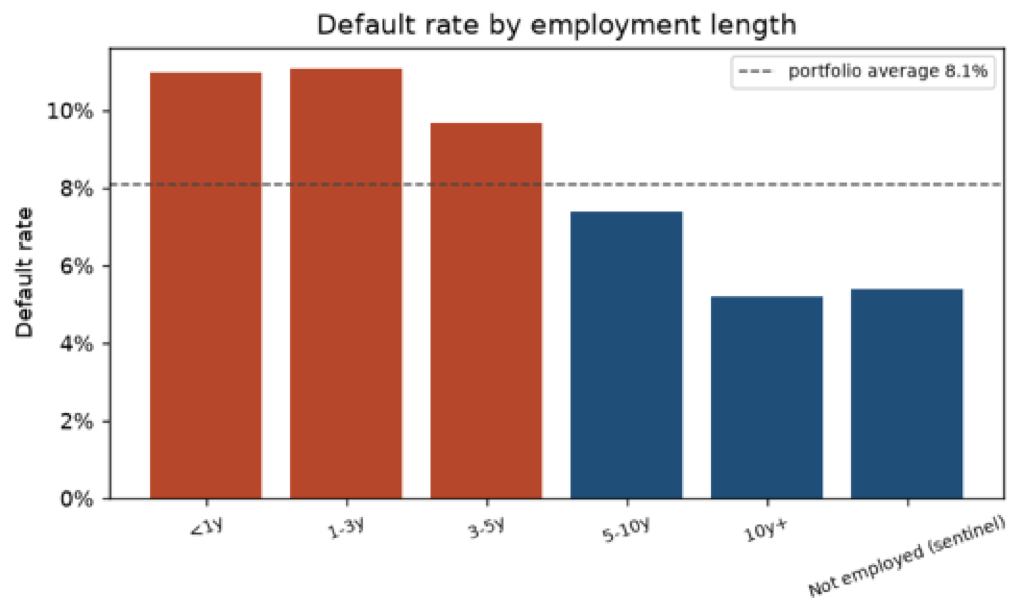


Dataset: Kaggle Home Credit Default Risk. Never committed to the repo.

EDA: the finding most analyses get wrong

DAYS_EMPLOYED = 365243 on 55,374 rows (18%) — a sentinel, not a 1000-year career

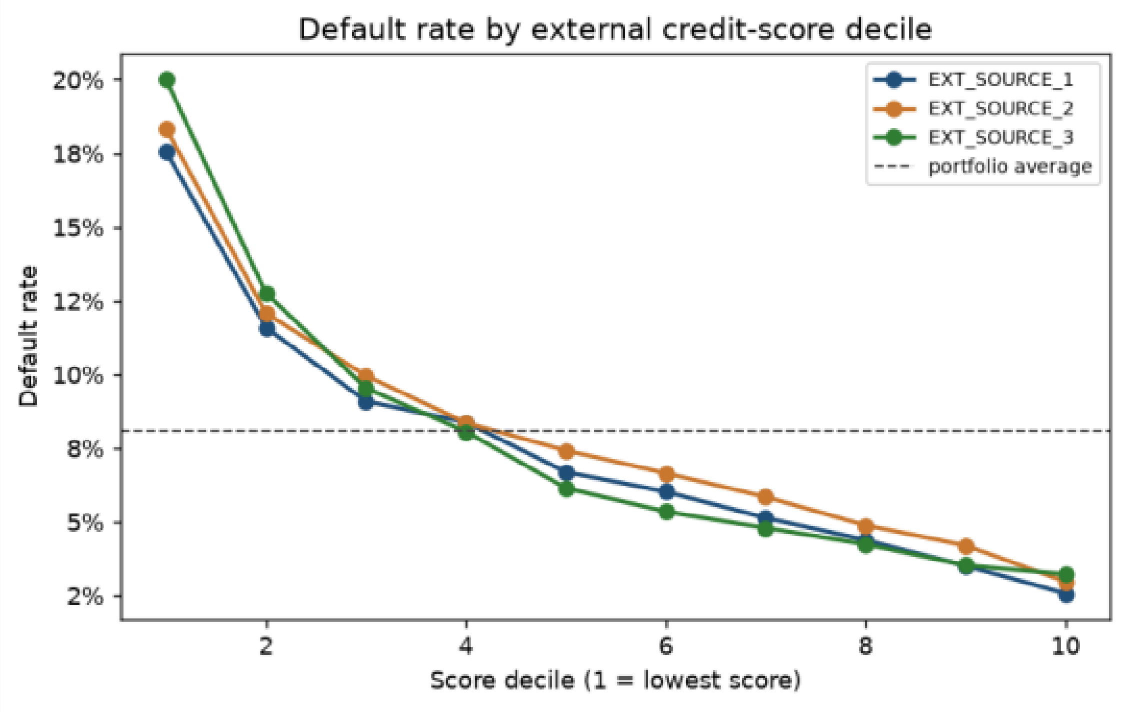
- Most write-ups stop at ‘dirty data, drop it’. But 99.96% of that group are pensioners.
- They default at 5.4% — BELOW the 8.7% employed average. One of the safer segments in the book.
- Bin them into ‘10 years+’ and the trend moves the wrong way for the wrong reason. Drop them and 18% of the portfolio disappears.
- Fixed in three places so it cannot leak back: the preprocessor, the SQL semantic layer, and the EDA itself.

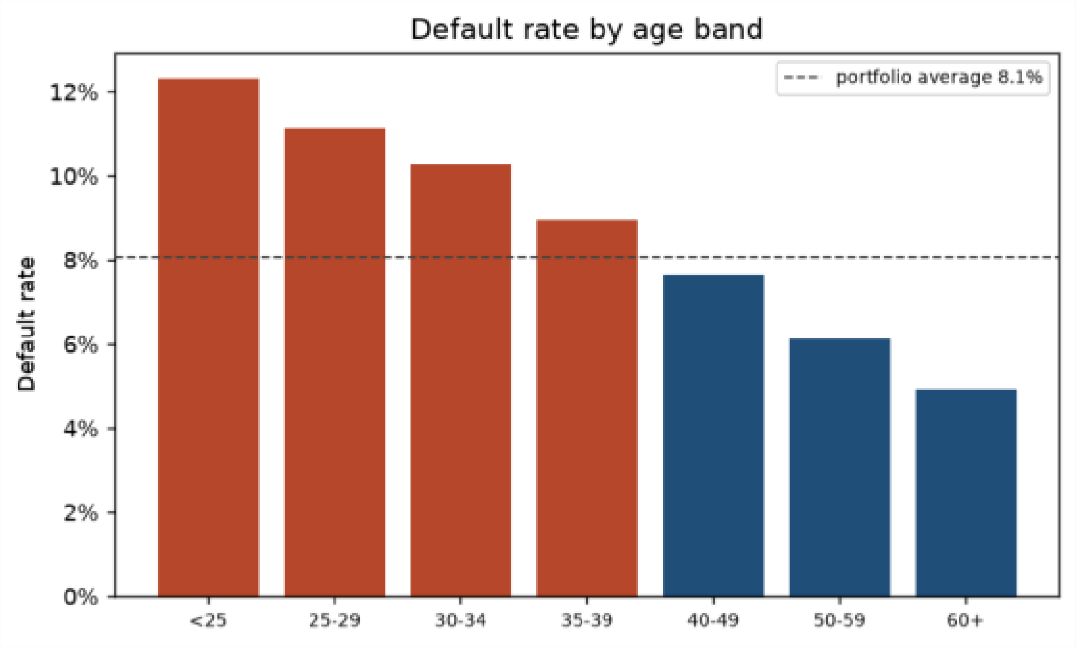


Four more insights in the notebook: income, contract type, age band, external scores.

EDA: where the signal actually lives

External credit scores separate the population ~6x, best decile to worst





Consequence: engineer mean / min / max / count / disagreement across the three scores — EXT_SOURCE_1 is missing for 56% of applicants.

The model, and the split design that keeps it honest

LightGBM, 143 features, four disjoint stratified splits

Split	Share	Used for
train	60%	Fitting the booster
valid	10%	Early stopping
calib	10%	Isotonic calibrator AND threshold selection
holdout	20%	Every headline number; the UI applicant pool

- Imbalance: scale_pos_weight = 11.39, not SMOTE.
- Calibration is not optional — reweighting wrecks the probabilities.

0.7788
ROC-AUC (holdout)

0.2637
PR-AUC

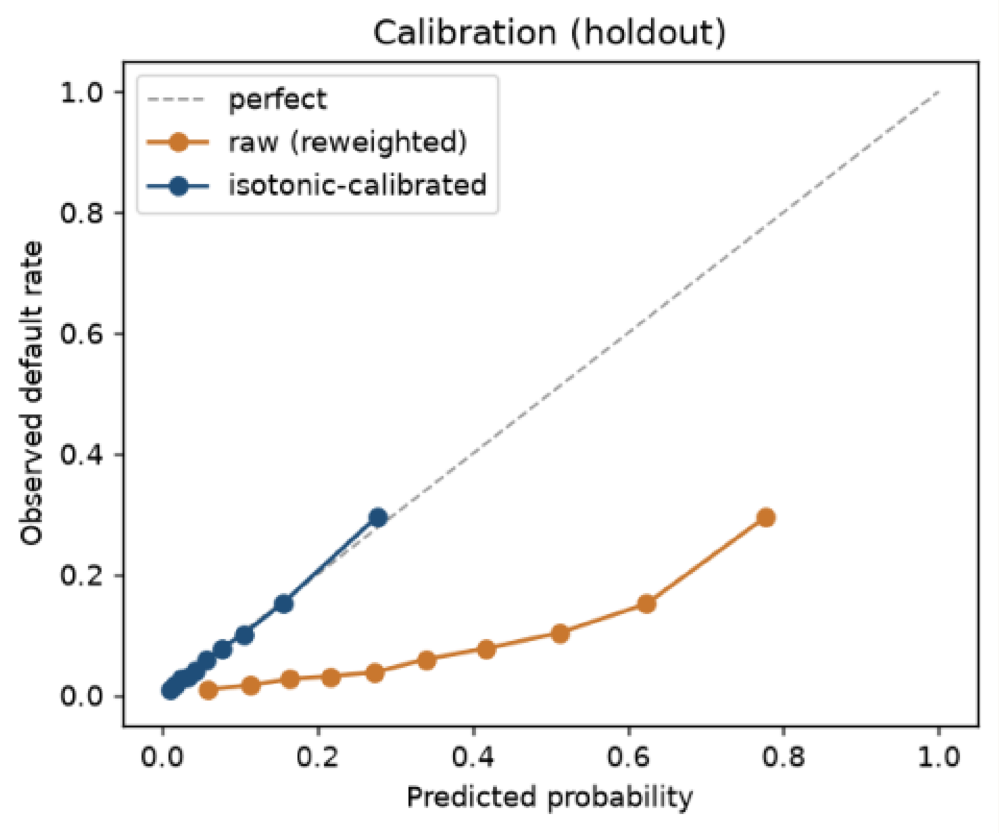
0.415
KS statistic

1190
Trees (early stopped)

Calibration: making the probability mean what it says

Raw model predicts 35% average risk against an actual 8.1%

- Brier score 0.1604 → 0.0663.
- Expected calibration error 0.2687 → 0.0039.
- Isotonic over Platt: the residual tail miscalibration is not sigmoid-shaped, and 30k calibration rows is ample for a non-parametric fit.
- Honest caveat: isotonic never reorders applicants, but it collapses 61k distinct scores onto ~150 steps. ROC-AUC moves 0.7792 → 0.7788.

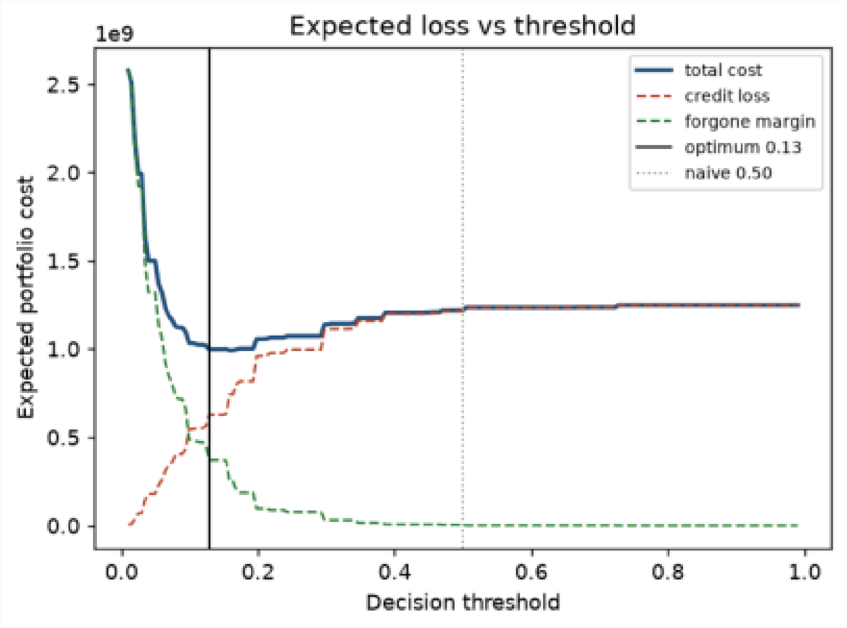


Fitted on a split used for nothing else — not the early-stopping set.

A threshold chosen on economics, not on 0.50

False negative costs 45% of the loan; false positive costs 8% — both weighted by that applicant's AMT_CREDIT

Policy	Threshold	Approvals	Expected loss
Approve everyone	—	100.0%	$1.25B
Naive 0.50	0.500	99.6%	$1.22B
Cost-optimal	0.129	81.4%	$998.42M



$221.39M of expected loss avoided versus naive 0.50.

$250.49M versus approving everyone.

Approval rate falls from 99.6% to 81.4% — the model buys that saving by declining 18 applicants in 100 it would otherwise have approved.

Threshold selected on the calibration split, reported on the holdout — so the saving is not the threshold overfitting its own evaluation set.

Decision Trace — the whole chain on one screen

The hero view of the app: one applicant, every stage, top to bottom

1. FEATURE ENGINEERING

143 features, 19 engineered, sentinel repaired

2. RISK MODEL

raw score 0.767 (LightGBM, scale_pos_weight 11.39)

3. CALIBRATION

24.2% calibrated probability of default

4. DECISION THRESHOLD

decline at or above 12.9% (cost-optimal)

5. RISK BAND

HIGH

6. EXPLANATION (SHAP → REASON CODES)

External credit scores are below the approved-applicant average

7. POLICY RULE

Average external credit score <= 0.26 AND Weakest external c... → DECLINE

Applicant 217620

DECLINE

24.2% probability of default

Principal reasons (ECOA-style)

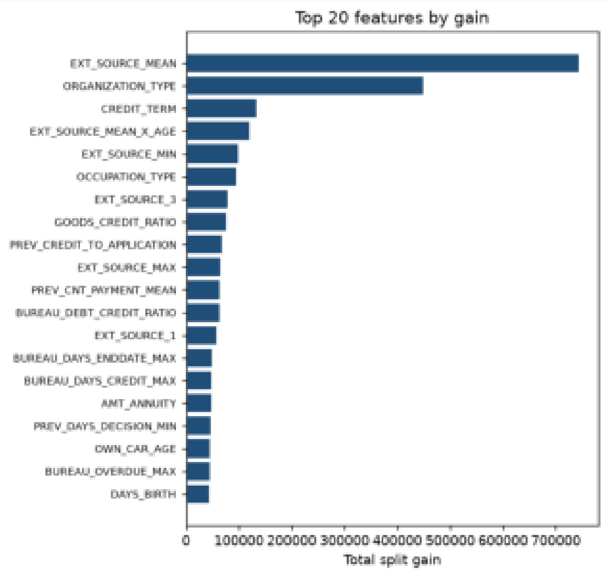
1. External credit scores are below the approved-applicant average
2. The implied repayment term is longer than typical
3. Employer industry is associated with higher default rates

Generated live from the trained model — not a mock-up.

Explainability that a regulator would accept

SHAP attribution → adverse-action reason codes (ECOA / Regulation B)

- Layer 1 — deterministic templates. Every reason code is derived from the SHAP contribution and the applicant's own value by rule: auditable, reproducible, works with no API key.
- Layer 2 — Claude Haiku restates those bullets as a short notice. It receives the reason codes as facts to restate, never as data to reason about.
- That ordering IS the hallucination control: the LLM is a stylist here, not a decision-maker.
- Protected attributes (gender, age, marital status) are suppressed from the notice and REPORTED — a reviewer sees that the model leaned on one, rather than having it quietly disappear.
- Stated caveat: SHAP explains the raw log-odds output, so contributions are shown as a share of total push, never as ‘+7% of default probability’.



From a 1,190-tree ensemble to credit policy

Depth-4 decision-tree surrogate of the model's own decisions — 81.2% fidelity

Rule (abbreviated)	Support	Default rate	Lift
Average external credit score <= 0.26 AND Weakest external ...	5.1%	28.2%	3.50x
Average external credit score > 0.26 AND Average external ...	4.5%	18.8%	2.33x
Average external credit score <= 0.36 AND Weakest external ...	3.6%	16.9%	2.09x
Average external credit score > 0.36 AND Average external ...	5.1%	13.6%	1.68x

- Precision is measured against ground-truth outcomes, not model predictions — so a rule is defensible on its own evidence.
- Rule language excludes categorical splits (an encoding artefact) and protected attributes — this is the one artefact a human might apply by hand.

14 rules total, partitioning the holdout population. Portfolio base rate 8.06%.

Talk to Data: self-correcting NL → SQL

Claude Sonnet 5 writes the SQL; guardrails decide whether it runs

```
Sonnet (schema + metrics + exemplars, cached)
   → run_sql(sql)
        → rejected or SQLite error → error text back to Sonnet → rewrite (≤ 2 retries)
        → success → rows to Haiku → business answer
```

#	Guardrail	Stops
1	Statement allowlist (sqlparse)	Anything that is not a single SELECT / WITH
2	Whole-token keyword denylist	INSERT / UPDATE / DELETE / DROP / ATTACH / PRAGMA
3	Table allowlist	Hallucinated tables, cross-database reads
4	Read-only connection (mode=ro)	Any write that survived layers 1-3

- Semantic layer: 15 metrics pinned once and injected into every prompt, so ‘default rate’ is one formula — and the 365243 sentinel is baked into employment_years, so the metric cannot lie.

25 labelled eval cases including a multi-turn follow-up, three unanswerable questions and a prompt-injection attempt.

Model health & fairness, labelled honestly

A model can hold 0.78 AUC overall while approving one group very differently

Group	Applicants	Default rate	ROC-AUC	Approval rate
gender: F	40,561	7.0%	0.772	85.0%
gender: M	20,940	10.2%	0.778	74.3%
age band: 45-59	20,645	6.9%	0.769	86.7%
age band: 35-44	16,893	8.3%	0.779	80.8%
age band: 25-34	14,325	10.5%	0.778	71.0%

Demographic parity gap: 10.7% by gender, 30.5% by age band.

None of these are ‘fixed’ here. Which parity a lender should enforce is a policy and legal question, not a modelling one — the platform measures them and puts them on screen.

Distribution shift is reported as train-vs-holdout PSI and explicitly NOT called drift: this dataset has no time axis, so max PSI is 0.0004 by construction. Calling that ‘no drift detected’ would be theatre.

Engineering

One container, mounted data, turnkey first run

```
data/raw/*.csv (mounted)
   → loader + preprocessor      sentinel repair, 19 engineered features
   → train.py                  LightGBM + isotonic + cost threshold
   → shap_explainer / reason_codes / rule_extractor
   → build_database            curated SQLite warehouse (read-only)
   → nl_to_sql + eval_harness  self-correcting, measured
   → agent/orchestrator        Claude tool-use over all six tools
   → app/ui.py                 7 sections, Decision Trace hero
```

- docker compose up builds the warehouse, EDA, model, rules and fairness report on first start, then serves the UI. Verified cold-start: 85 seconds.
- 121 passing tests. Guardrails, semantic-layer SQL executed against the real database, preprocessing traps and reason-code compliance all covered.
- Every LLM feature has a deterministic twin — scoring, explanation, rules and EDA all run with no API key.
- Token discipline, measured: 18,653 → 5,027 billable input tokens over a 6-turn session (−73%), driven mostly by prompt caching.

What makes this a system, not a notebook

— Calibrated probabilities, not just a ranking.

— A threshold chosen on expected loss, not on 0.50.

— Reasons a regulator would accept, generated without an LLM in the loop.

— Policy rules with support, precision and lift attached.

— SQL a language model cannot misuse, on four independent guardrails.

— Fairness gaps measured and published rather than absorbed.

— Limitations stated: no time axis, an assumed cost matrix, 81% surrogate fidelity.

Every number in this deck regenerates from the pipeline: python documents/build_presentation.py